HOW CONTEXT HELPS: A DISCRIMINATIVE CODEWORD SELECTION METHOD FOR OBJECT DETECTION

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ABSTRACT

We first propose in this paper to localize objects in images based on the models learned from the weakly labeled images. This task is termed as region of interest (ROI) detection. Local features such as SIFT or HOG are extracted and the discriminative words from clustered codewords based on SIFT and HOG are selected to model the objects. Then how to find the discriminative words to model the object is important. Existing ROI detection methods consider the information from the foreground objects by selecting the words appearing more in the images belonging to one specific image class. Considering the information from background/context is also helpful for object detection and classification, we propose to select the discriminative words which appear more in the foreground/object and less in the background/context.

Second, another task is to give the class label (object in this setting) for a given image and also give the position of the object appearing in the image. This task is termed as objection detection. A normal way for this task after ROI is to extract features from the detected regions and not from the whole image. Since the discriminative words extracted during ROI detection has good discriminative ability, we propose to use these words for object detection. Experimental results on PASCAL VOC 2006 dataset and a larger dataset containing 29 classes demonstrate the effectiveness of the proposed method.

Index Terms— Region of interest, bag of features, object and context, discriminative words, localization, detection

1. INTRODUCTION

In recent years, great improvements have been made in object classification and detection. Related works [1, 2, 3, 4, 5] and some works on evaluation competitions such as PASCAL Visual Object Classes Challenge (VOC) [6] and TRECVid [7], demonstrate good performance of concept classification and detection. State-of-the-art methods are based on the bag-of-feature (BOF) model [1, 8]. Specifically, visual features of local keypoints (e.g. SIFT [9]) are first extracted from images. Then the extracted local features are quantized into a set of visual codewords and histogram is generating by estimating the frequency of the codewords to represent the image.

The relationship between concepts and codewords has been widely discussed. A quantizer codebook learning algorithm is proposed in [10], which considers the posterior distribution of concepts in the codebook learning step. On the other hand, [11] proposes a feature selection approach which gives each codeword a confidence for each concept. We extend this work by considering the information from both foreground and background based on the predicted results.

In this paper, we study the problem of object localization and detection with foreground and context information. Specifically, object localization is to give the position of one object appearing in the image given the image class. And object detection is to give the class label of a given image and also give the position of the object appearing in the image. In particular, we are interested in methods that allow a small number of object annotation, i.e., with bounding box annotation of object appearing in the image, to build the contextual model for improving object localization, and help improving the object detection results.

The focus of our work is to learn the contextual model. In [11], examplar model for region-of-interest (ROI) detection is proposed. This method does not detect a saliency map of image from a bottom-up way [12], but learn the ROI models by selecting the discriminative local features. Uijlings *et al.*'s object (foreground)/context (background) modeling work shows how the context information is important in a BOF framework for the large and realistic dataset [13]. Inspired by this work, we propose a new discriminative codeword selection criterion which considers information from both object and context for object localization and detection.

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2. CONTEXTUAL MODEL FOR OBJECT LOCALIZATION AND DETECTION

Our proposed contextual model for object localization and detection includes two phases of training and test. The training phase is shown in Fig. 1 and will be introduced in Section 2.1. The trained model can be used for object localization and detection, which will be introduced in Section 2.2 and 2.3, respectively.

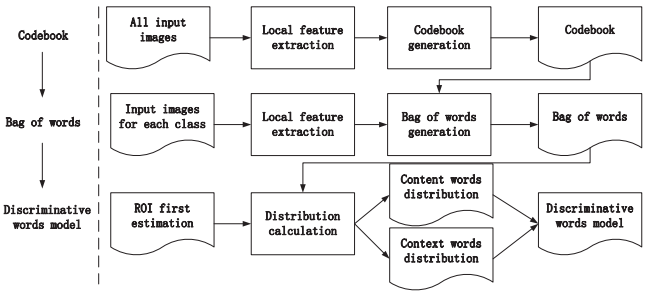

Fig. 1. Training phase of our proposed contextual model.

2.1. Training of Our Proposed Contextual Model

The goal of our proposed contextual modeling method is to learn a model that can describe each kind of objects. Specifically, the input for the training phase are a small set of images in which some specific class of objects appearing and with bounding box to show the position of object appearing in the image. And a set of visual words are selected to represent the object class. Then each image can be represented as histogram of these words. In [11], the distribution of the words appearing in one specific class of images are computed, and the words having highest values are selected. Equ. (1) shows the function $D(w_i)$ to select discriminative words w_i.

$$D(w_i) \sim \frac{N_i}{N} \tag{1}$$

where N_i denotes the number of class labeled images containing w_i, N the number of images in database containing w_i. Then the key points having the words same to those discriminative words are kept. And the bounding box with the points inside forms the initial prediction of the class instance appearing in the image. This method selects words which appearing more in one specific class of images. However, for one specific class, the words will come from both object and context part (if the context part also appearing altogether in all these training images belonging to one specific class). For example, normally "grass" appears together with "tiger". Then we propose to select discriminative words which appear more in object part and less in context part.

Based on the first estimation of ROI detection, the words corresponding to keypoints inside the predicted bounding box are defined as object (O), and that outside the predicted bounding box are defined as context (C). The definition is similar to that as in [13]. Our proposed discriminative words selection function is given in Equ. (2).

$$D(w_i) = \frac{|O_i - C_i|^2}{O_i + C_i} \cdot \text{sign}(O_i - C_i) \tag{2}$$

where O_i and C_i are the distribution of word w_i in object and context, respectively. If $D(w_i)$ is high, the word w_i is more likely to appear in object than that appear in context. We sort the words according to $D(w_i)$ values and choose the top n words to form the discriminative word set for one specific object class.

2.2. Contextual Model for Object Localization

After the discriminative words for one specific object class is learned, they can be used for object location. Specifically, for a testing image belonging to this class, local features are extracted from the image. Each local feature is assigned to one of the words in codebook. And the points having words same to those discriminative words are kept. Then the bounding box of these kept points is formed to indicate the position of the object appearing in the image.

2.3. Contextual Model for Object Detection

In [11], for object classification, the codebook histogram within the predicted ROI region for one image class is computed. For training the classifier such as Support Vector Machine (SVM), positive samples are from points from within the ROI belonging to one image class, and negative samples are from points within the ROI belonging to other classes. To incorporate more spatial information, three-level spatial pyramid matching (SPM) method [3] is used for comparing the image ROIs in 1×1, 2×2, and 4×4 together.

For a given test image, all the trained classes are tested separately. The final classification result is the class that gives the highest classifier confidence. Furthermore, only the evaluation measure of localization is larger than a threshold is considered as a correct detection for one specific class.

3. EXPERIMENTS

The performance of the proposed method is evaluated on two datasets. The first dataset is PASCAL VOC2006 dataset [6]. In this dataset, 6 classes of images are used. The classes are bicycle, bus, car, motorbike, cow, and sheep. Considering different viewpoints of one class (such as "bicycleLeft" and "bicycleRight"), 15 categories are defined. Each category has 15-20 images for training and another 15-20 images for test. Fig. 2 gives some example images of VOC2006.

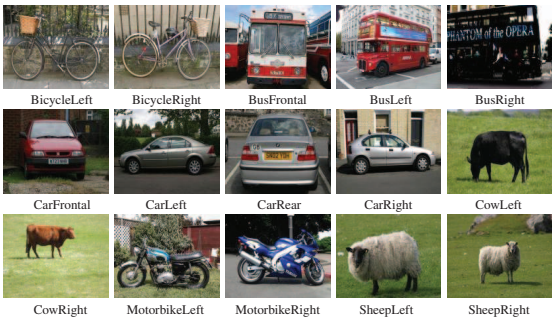

Fig. 2. Example images of PASCAL VOC2006 dataset.

We also extend the 6 classes VOC2006 dataset to 29 classes for the localization task. Considering different viewpoints, there are 41 categories in total. [1] Each category has 25 images for training and another 25 images for test.

3.1. Experiments for Localization Task

For localization, we use Area of Overlap (AO) to evaluate the performance [6]. The AO is defined as follows:

$$AO(B_{gt}, B_p) = \frac{|B_{gt} \cap B_p|}{|B_{gt} \cup B_p|} \tag{3}$$

where B_{gt} is the ground-truth bounding box by annotator, B_p is the bounding box given by our localization method, $|B_{gt} \cap B_p|$ is the area of the intersection of the two bounding boxes and $|B_{gt} \cup B_p|$ is the area of the union of the two bounding boxes. Higher AO indicates better localization performance.

3.1.1. Experiments on VOC2006 Dataset

In our experiment, local features are extracted by Harris-Laplace detector [14] and SIFT descriptor [9] for describing the points and their neighboring information. The codebook is generated by k-means clustering algorithm. k is set to 3000 as in [11]. For each category, 64 discriminative words are selected from the codebook. Fig. 3 shows the experimental results on VOC2006 dataset by method in [11] and our proposed contextual model.

It can be observed from Fig. 3 that our proposed contextual model can obtain better localization method. The mean AO of the method in [11] is 0.50, and our proposed contextual model can obtain mean AO of 0.66.

Fig. 4 shows some examples of the results. It can be observed that points corresponding to words appearing in context are eliminated by using our proposed contextual method.

[1] The 29 classes and 41 categories are: banana, bicycle (left, right), bookcase, bus (frontal, left, right), butterfly, car (frontal, left, rear, right), cat, cow (left, right), dog, dolphin, elephant, lion (male, female), flower, giraffe, goose, guitar, horse, jackfruit, kangaroo, ketch, lamp, motorbike (left, right), penguin, person, piano, plane (left, right), pomegranate, sheep (left, right), sideboard, tiger.

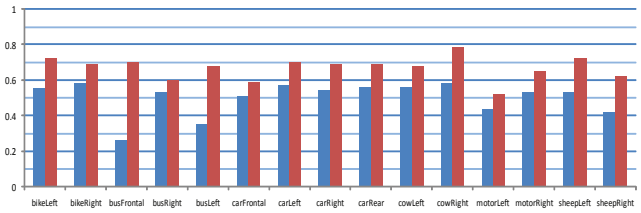

Fig. 3. Localization performance in VOC2006 dataset. For each category, the blue bin is the AO value by using method of [11], and the red bin is that by our proposed contextual model.

Fig. 4. Example images of the object localization results. The top row is the result by using method of [11]. The bottom row is result by using our proposed contextual model.

3.1.2. Experiments on 41-Categories Dataset

We also do experiment on extended 41-categories dataset. In which, we construct the codebook with 20,000 words. The number of the discriminative words is set to 512. Fig. 5 shows the experimental result.

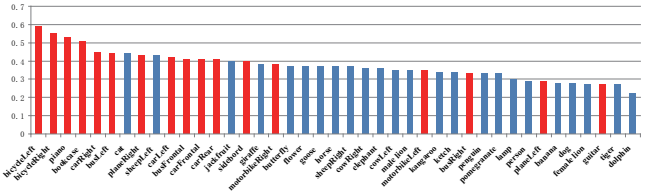

Fig. 5. Localization performance on 41-categories dataset. The red bins are the AO values of rigid object categories, and the blue bins are that of non-rigid object categories.

It can be observed from Fig. 5 that the object localization for most rigid object categories outperforms that for non-rigid object categories. Furthermore, the mean AO of the object localization for extended 41-catetories dataset is 0.38, which is lower than that for VOC2006 dataset. The object localization performance is decreased with the number of categories increased. To describe more variations on more categories, although we have changed the codebook size from 3,000 to 20,000, the fixed number of discriminative words may not be proper for 41-catetories dataset. Then another experiment is performed to explore the influence of the size of discrimina-

tive words. For codebook size of 20,000, experiments are done on changing the number of discriminative words from 64 to 768. The experimental results are shown in Fig. 6.

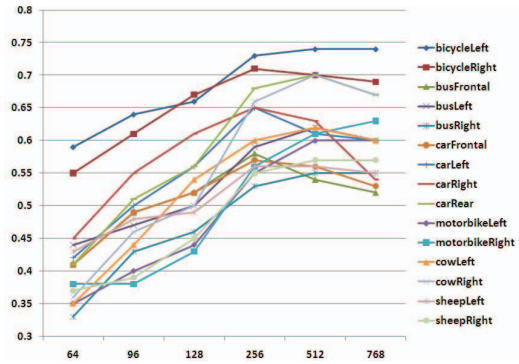

Fig. 6. Localization performance by using different number of discriminative words for 41-catetories dataset.

It can be observed from Fig. 6 that the localization performance doesnot always increased with more words used. Furthermore, the proper size of discriminative words may not be same for different categories. Considering all the classes in our experiment have a accepted localization performance, it suggests a words size of 2% of the codebook size.

3.2. Experiments for Detection Task

We use VOC2006 dataset to evaluate object detection based on our proposed contextual model. Fig. 7 shows the object detection performance by using the method in [11] and our proposed contextual model. The Average Precision (AP) measure of the detection results are tabulated in Table 1.

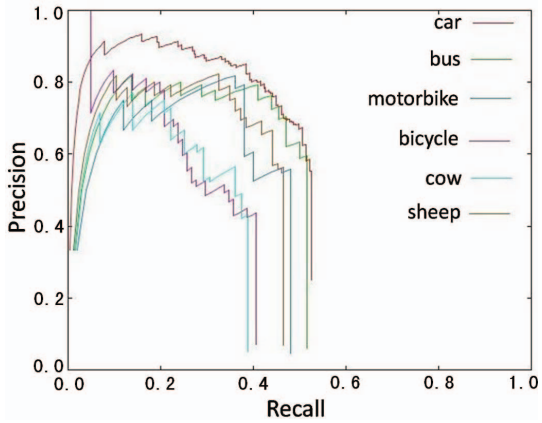

Fig. 7. Object detection performance by using the method in [11] and our proposed contextual model.

It can be observed from Fig. 7 that our proposed contextual model can obtain promising performance of object detection for VOC2006 dataset. Comparison results in Table 1 shown that among the six classes tested, our proposed contextual model can obtain improved performance. And for the two classes (motorbike and bicycle), our proposed model can obtain comparable performance to that by the method in [11]. The mean Average Precision (mAP) is improved from 0.320 to 0.334.

Table 1. Average precision(AP) of object detection by using the method in [11] and our proposed contextual model.

	Car	Bus	Motor.	Bic.	Cow	Sheep	Mean
[11]	0.412	0.249	**0.371**	**0.482**	0.212	0.195	0.320
Ours	**0.436**	**0.375**	0.326	0.466	**0.249**	**0.239**	**0.334**

Furthermore, it can be observed that the detection performance for rigid object classes (e.g., car, bus, motorbike, bicycle) is higher that that for non-rigid object classes (e.g., cow, sheep).

4. CONCLUSION

We proposed in this paper a novel method to select discriminative words for object localization and object detection. Experimental results on VOC2006 dataset demonstrate that by considering the information from both object and context, the object can be better localized and detected. Also, experimental results on extended dataset from VOC2006 dataset, i.e. 41-categories dataset, show the method performs better for rigid object classes than non-rigid object classes. Our future work is to explore more effective features for both rigid and non-rigid object classes.

5. REFERENCES

[1] J. Sivic and A. Zisserman, "Video google: a text retrieval approach to object matching in videos," in *ICCV*, 2003.

[2] Fei-Fei Li and Pietro Perona, "A bayesian hierarchical model for learning natural scene categories," in *CVPR*, 2005.

[3] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *CVPR*, 2006.

[4] J. Zhang, M. Marszalek, S. Lazebnik, and C. Schmid, "Local features and kernels for classification of texture and object categories: A comprehensive study," *IJCV*, 2007.

[5] K. van de Sande, T. Gevers, and C. Snoek, "Evaluation of color descriptors for object and scene recognition," in *CVPR*, 2008.

[6] M. Everingham, A. Zisserman, C. K. I. Williams, and L. Van Gool, "The PASCAL Visual Object Classes Challenge 2006 (VOC2006) Results," http://www.pascal-network.org/challenges/VOC/voc2006/results.pdf.

[7] A. F. Smeaton, P. Over, and W. Kraaij, "Evaluation campaigns and TRECVid," in *MIR*, 2006.

[8] G. Csurka, C. R. Dance, L. Fan, J. Willamowski, and C. Bray, "Visual categorization with bags of keypoints," in *ECCV Workshop on Statistical Learning in Computer Vision*, 2004.

[9] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *IJCV*, 2004.

[10] Svetlana Lazebnik and Maxim Raginsky, "Supervised learning of quantizer codebooks by information loss minimization," *IEEE Trans. PAMI*, vol. 31, no. 7, pp. 1294–1309, 2009.

[11] O. Chum and A. Zisserman, "An examplar model for learning object classes," in *CVPR*, 2007.

[12] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *TPAMI*, 1998.

[13] J.R.R. Uijlings, A.W.M. Smeulders, and R.J.H. Scha, "What is the spatial extent of an object?," in *CVPR*, 2009.

[14] K. Mikolajczyk and C. Schmid, "Scale & affine invariant interest point detectors," *IJCV*, 2004.